

Type text to search here...

Wireless Quiz

Go to comments

Result of Wireless Quiz:

Total Questions	Full Score	Passing Rate	Your Score	Correct Answer Percentage	Elapsed
13	160	80%	160	100%	00:01:13

Congratulations!

You passed this test!

If you want to retake this quiz, please press Ctrl + F5 on Windows or press CMD + R on Mac.

Your answers are shown below:

Question 1

Which 802.11 frame type is association response?

- A. protected frame
- B. action
- C. control
- **D. management**

Explanation

There are three main types of 802.11 frames: the Data Frame, the Management Frame and the Control Frame. Association Response belongs to Management Frame. Association response is sent in response to an association request.

Question 2

Which statement about Link Aggregation when implemented on a Cisco Wireless LAN Controller is true?

- A. When enabled the WLC bandwidth drops to 500 Mbps
- B. The EtherChannel must be configured in “mode active”
- **C. One functional physical port is needed to pass client traffic**
- D. To pass client traffic two or more ports must be configured

Explanation

Link aggregation (LAG) is a partial implementation of the 802.3ad port aggregation standard. It bundles all of the controller’s distribution system ports into a single 802.3ad port channel.

Restriction for Link aggregation:

+ LAG requires the EtherChannel to be configured for ‘mode on’ on both the controller and the Catalyst switch -> Answer 'The EtherChannel must be configured in “mode active”' is not correct.
+ If the recommended load-balancing method cannot be configured on the Catalyst switch, **then configure the LAG connection as a single member link** or disable LAG on the controller -> Answer 'To pass client traffic two or more ports must be configured' is not correct while answer 'One functional physical port is needed to pass client traffic' is correct.

Reference: https://www.cisco.com/c/en/us/td/docs/wireless/controller/7-5/configuration-guide/b_cg75/b_cg75_chapter_0100010.html

Question 3

Which mode allows access points to be managed by Cisco Wireless LAN Controllers?

- **A. lightweight**
- B. mobility express
- C. bridge
- D. autonomous

Explanation

A Lightweight Access Point (LAP) is an AP that is designed to be connected to a wireless LAN (WLAN) controller (WLC). APs are “lightweight,” which means that they cannot act independently of a wireless LAN controller (WLC). The WLC manages the AP configurations and firmware. The APs are “zero touch” deployed, and individual configuration of APs is not necessary.

Reference: https://www.cisco.com/c/en/us/support/docs/wireless/aironet-1200-series/70278-lap-faq.html

Question 4

Several new coverage cells are required to improve the Wi-Fi network of an organization. Which two standard designs are recommended? (Choose two)

- A. 5GHz channel selection requires an autonomous access point.
- B. Adjacent cells with overlapping channels use a repeater access point.
- **C. 5GHz provides increased network capacity with up to 23 nonoverlapping channels.**
- **D. Cells that overlap one another are configured to use nonoverlapping channels.**
- E. For maximum throughput, the WLC is configured to dynamically set adjacent access points to the same channel.

Explanation

The 802.11 standard defines 23 20MHz wide channels in the 5 GHz spectrum. Each channel is spaced 20MHz apart and separated into three Unlicensed National Information Infrastructure (UNII) bands.

Reference: https://documentation.meraki.com/MR/WiFi_Basics_and_Best_Practices/Channel_Planning_Best_Practices

Question 5

Which QoS Profile is selected in the GUI when configuring a voice over WLAN deployment?

- A. Gold
- B. Silver
- **C. Platinum**
- D. Bronze

Explanation

Cisco Unified Wireless Network solution WLANs support four levels of QoS: Platinum/Voice, Gold/Video, Silver/Best Effort (default), and Bronze/Background.

Reference: https://www.cisco.com/c/en/us/td/docs/wireless/controller/7-4/configuration/guides/consolidated/b_cg74_CONSOLIDATED/b_cg74_CONSOLIDATED_chapter_01010111.html

Question 6

Which type of wireless encryption is used for WPA2 in pre-shared key mode?

- A. AES-256
- B. RC4
- C. TKIP with RC4
- **D. AES-128**

Explanation

WPA 2 implements the National Institute of Standards and Technology (NIST)-recommended Advanced Encryption Standard (AES) encryption algorithm with the use of Counter Mode with Cipher Block Chaining Message Authentication Code Protocol (CCMP). AES Counter Mode is a block cipher that encrypts 128-bit blocks of data at a time with a 128-bit encryption key.

Reference: https://www.cisco.com/c/en/us/support/docs/wireless-mobility/wireless-lan-wlan/67134-wpa2-config.html

Also here is a quick comparison between WPA, WPA2 & WPA3:

	WEP	WPA	WPA2	WPA3
Released	1997	2003	2004	2018
Encryption protocol	RC4	TKIP	CCMP (with AES)	GCMP (with AE
Session Key	40 or 104bits	128bits	128bits	128/256bits
Authentication	Open system (OSA)/ Shared Key (SRA)	802.1X / Pre-shared Key	802.1X / Pre-shared Key	SAE (Simultane Authentication of E
Level of security	Low	Medium	High	Very High
Vulnarabilities	Easily cracked due to weak RC4	Vulnerable to TKIP exploits	PSK vulnerable to brute force	Improved against I Brute Force and I attacks

Reference: https://www.networkacademy.io/ccna/wireless/client-authentication-methods

Question 7

Which two values or settings must be entered when configuring a new WLAN in the Cisco Wireless LAN Controller GUI? (Choose two)

- **A. Profile name**
- B. QoS settings
- C. ip address of one or more access points
- D. management interface settings
- **E. SSID**

Question 8

Which feature on the Cisco Wireless LAN Controller when enabled restricts management access from specific networks?

- A. Flex ACL
- **B. CPU ACL**
- C. RADIUS
- D. TACACS

Explanation

Whenever you want to control which devices can talk to the main CPU, a CPU ACL is used.

Note: CPU ACLs only filter traffic towards the CPU, and not any traffic exiting or generated by the CPU.

Reference: https://www.cisco.com/c/en/us/support/docs/wireless/4400-series-wireless-lan-controllers/109669-secure-wlc.html

Question 9

What is a benefit of using a Cisco Wireless LAN Controller?

- A. Central AP management requires more complex configurations
- **B. It eliminates the need to configure each access point individually**
- C. Unique SSIDs cannot use the same authentication method
- D. It supports autonomous and lightweight APs

Question 10

What is a design element when deploying an 802.11b wireless infrastructure?

- A. setting the maximum data rate to 54 Mbps on the Cisco Wireless LAN Controller
- B. disabling TPC so that access points can negotiate signal levels with their attached wireless devices.
- C. configuring access points to provide clients with a maximum of 5 Mbps
- **D. allocating non overlapping channels to access points that are in close physical proximity to one another**

Question 11

When configuring a WLAN with WPA2 PSK in the Cisco Wireless LAN Controller GUI, which two formats are available to select? (Choose two)

- A. binary
- B. base64
- **C. hexadecimal**
- D. decimal
- **E. ASCII**

Explanation

When configuring a WLAN with WPA2 Preshared Key (PSK), we can choose the encryption key format as either ASCII or HEX.

Reference: https://www.cisco.com/c/en/us/td/docs/wireless/controller/9800/config-guide/b_wl_16_10_cg/multi-preshared-key.pdf

Question 12

Which unified access point mode continues to serve wireless clients after losing connectivity to the Cisco Wireless LAN Controller?

- **A. flex connect**
- B. mesh
- C. local
- D. sniffer

Explanation

In previous releases, whenever a FlexConnect access point disassociates from a controller, it moves to the standalone mode. The clients that are centrally switched are disassociated. However, the FlexConnect access point continues to serve locally switched clients. When the FlexConnect access point rejoins the controller (or a standby controller), all clients are disconnected and are authenticated again. This functionality has been enhanced and the connection between the clients and the FlexConnect access points are maintained intact and the clients experience seamless connectivity. When both the access point and the controller have the same configuration, the connection between the clients and APs is maintained.

Reference: https://www.cisco.com/c/en/us/td/docs/wireless/controller/7-4/configuration/guides/consolidated/b_cg74_CONSOLIDATED/b_cg74_CONSOLIDATED_chapter_010001101.html

Question 13

An engineer must configure a WLAN using the strongest encryption type for WPA2-PSK. Which cipher fulfills the configuration requirement?

- **A. AES**
- B. RC4
- C. WEP
- D. TKIP

Explanation

Many routers provide WPA2-PSK (TKIP), WPA2-PSK (AES), and WPA2-PSK (TKIP/AES) as options. TKIP is actually an older encryption protocol introduced with WPA to replace the very-insecure WEP encryption at the time. TKIP is actually quite similar to WEP encryption. TKIP is no longer considered secure, and is now deprecated. In other words, you shouldn’t be using it.

AES is a more secure encryption protocol introduced with WPA2 and it is currently the strongest encryption type for WPA2-PSK.

Comments (15) Comments

1. Anonymous
   August 12th, 2020

   cool

2. Joe
   April 9th, 2021

   the response for the Q12 is wrong in the quiz, the correct answers are ASCII and base64 not hexadecimal.

3. anon
   May 30th, 2021

   @Joe are you sure about that?

4. Jacob B
   April 19th, 2022

   @Joe is wrong. Refer to Step 8: https://www.cisco.com/c/en/us/td/docs/wireless/controller/7-4/configuration/guides/consolidated/b_cg74_CONSOLIDATED/b_cg74_CONSOLIDATED_chapter_01010001.html

5. Anonym
   June 22nd, 2022

   Hello! i think the question about WPA2 encryption method is wrong because WPA2 generates 256 bits of keying material — the “Pairwise Master Key” (PMK) — and derives different keys from it, one of which is the 128-bit “Pairwise Temporal Key” (PTK) used for AES-128 encrypting the traffic between that one client and that one AP for that one session. So I believe the correct answer is 128-bit key which is written and on the book of “CCNA CERTIFICATION STUDY GUIDE EXAM 200-301 by Todd Lamie”

6. Beka
   January 25th, 2023

   @Anonym, you are correct, The type of wireless encryption used for WPA2 in pre-shared key mode is:
   D. AES-128

7. Solaire
   March 15th, 2023

   @9tut regarding WPA2 PSK – AES-128 is used, as mentioned by the two people before me. Pls fix answer

8. Pacas
   September 27th, 2023

   @9tut Hi could you please check this post?

   regarding WPA2 PSK – AES-128 is used, as mentioned by the two people before me. Pls fix answer

9. Pinza
   May 20th, 2024

   I think that for the question :
   Which feature on the Cisco Wireless LAN Controller when enabled restricts management access from specific networks?
   The correct is FLEX ACL

10. CCNA Student
    October 12th, 2024

    @9tut,

    Regarding the question:
    “Which type of wireless encryption is used for WPA2 in pre-shared key mode?”, it appears there may be a mistake with the answer given?

    The link that was provided in the answer states:

    “AES Counter Mode is a block cipher that encrypts 128-bit blocks of data at a time with a 128-bit encryption key.”

    I didn’t find anywhere in the article that mentions 256.

11. 9tut
    October 13th, 2024

    @CCNA Student: We changed the explanation for this question, please check again.

12. User1
November 15th, 2024

“Which type of wireless encryption is used for WPA2 in pre-shared key mode?”

Should be AES-128.

WPA3 offers AES-128 in personal mode and AES-192 in Enterprise mode. WPA2 offers AES-128 in both personal (PSK) and enterprise mode.

13. NowHearDis
April 13th, 2026

@9TUT Your reference link is wrong. Try referencing CISCO here:
https://www.cisco.com/c/en/us/support/docs/wireless-mobility/wireless-lan-wlan/67134-wpa2-config.html

Which type of wireless encryption is used for WPA2 in pre-shared key mode?

A. RC4
B. TKIP with RC4
C. AES-128
D. AES-256

The correct answer is AES-128.

“WPA 2 implements the National Institute of Standards and Technology (NIST)-recommended Advanced Encryption Standard (AES) encryption algorithm with the use of Counter Mode with Cipher Block Chaining Message Authentication Code Protocol (CCMP). AES Counter Mode is a block cipher that encrypts 128-bit blocks of data at a time with a 128-bit encryption key.”

14. 9tut
April 14th, 2026

@User1, @NowHearDis: Thank you for your information, we updated it!

15. @9tut Dead explanation related links
July 14th, 2026

Hello,
Question’s 1,2,3,10 explanation links are pointing to wrong sites.
Please correct them
Thank you very much

Add a Comment
Name

Submit Comment
Subscribe to comments feed
SDN Quiz IPv6 Quiz

Premium Member Zone

Welcome Darène MAYISSA!

- Welcome Premium Member!
- CCNA - New Questions Part 5
- CCNA - New Questions Part 6
- CCNA - New Questions Part 7

- CCNA - New Questions Part 8
- CCNA - New Questions Part 9
- CCNA - New Questions Part 10
- CCNA - New Questions Part 11
- CCNA - New Questions Part 12
- CCNA - New Questions Part 13
- CCNA - New Questions Part 14
- CCNA - New Questions Part 15
- CCNA - New Questions Part 16
- Composite Quizzes
- === New CCNA v1.1 Lab Sims (very important) ===
- Voice VLAN Configuration Sim 2
- Voice VLAN & LLDP Sim
- Allowed, Native VLAN & LACP Sim
- Static Routing Configuration Sim 4
- IPv4 and IPv6 Connectivity Sim
- VLAN & CDP Sim 2
- IPv4 and IPv6 Assignment Sim
- VLAN CDP & LLDP Sim
- OSPF Configuration Sim 2
- OSPF Configuration Sim 3
- Trunking & LACP Sim
- VLAN Configuration Sim
- Allowed, Native VLAN & LACP Sim 2
- 802.1Q Trunking & LACP
- IPv4 and IPv6 Assignment Sim 2
- 802.1Q Trunking, Native VLAN & LACP
- Static Routing Configuration Sim 5
- Static Routing Configuration Sim 6
- PAT, NTP & DHCP Relay Sim
- OSPF DR Sim
- NACL DHCP Snooping 3
- NAT DHCP SSH Sim
- ==== Lab Sims for CCNA (although they are rather old but please learn all of them too) ====
- IP Services Sim Version 2
- OSPF Configuration Sim
- LACP Configuration Sim
- Voice VLAN Configuration Sim
- VLAN and Trunking Configuration Sim
- Named Access-list & Port Security Sim
- Named Access-list & DHCP Snooping Sim
- VLAN and CDP Sim
- IPv4 and IPv6 Deployment Sim
- Static Routing Configuration Sim 3
- VLAN & LLDP Sim
- IPv6 Static Routing Sim
- Named Access-list & DHCP Snooping Sim 2
- ISL Trunking and LACP Sim
- === Below Sims are old and retired so you do not need to learn them ===
- Static Routing Configuration Sim 2
- IP Services Sim
- Static Routing Configuration Sim

CCNA Training

- CCNA Knowledge
- CCNA Tutorials, Practice Labs & Lab Challenges
- Spine-Leaf Architecture Tutorial
- Generative AI vs Predictive AI Tutorial
- REST API Tutorial
- Network Time Protocol (NTP) Tutorial
- DHCP Snooping Tutorial
- What is SDN?

- How a router makes a forwarding decision
- Configure SSH for Remote Access on Cisco Router
- Key Security Concepts Tutorial
- Forwarding per-hop behavior (PHB) for QoS Tutorial
- Port Security Tutorial
- Puppet Tutorial
- Chef Tutorial
- Ansible Tutorial
- JSON Tutorial
- Layer 2 Threats and Security Features
- AAA TACACS+ and RADIUS Tutorial
- STP Root Port Election Tutorial
- TCP and UDP Tutorial
- WAN Tutorial
- DHCP Tutorial
- Simple Network Management Protocol SNMP Tutorial
- Syslog Tutorial
- EtherChannel Tutorial
- Hot Standby Router Protocol HSRP Tutorial
- InterVLAN Routing Tutorial
- Cisco Command Line Interface CLI
- Cisco Router Boot Sequence Tutorial
- OSI Model Tutorial
- Subnetting Tutorial – Subnetting Made Easy
- Frame Relay Tutorial
- Wireless Tutorial
- Virtual Local Area Network VLAN Tutorial
- VLAN Trunking Protocol VTP Tutorial
- IPv6 Tutorial
- Rapid Spanning Tree Protocol RSTP Tutorial
- Spanning Tree Protocol STP Tutorial
- Network Address Translation NAT Tutorial
- Access List Tutorial
- RIP Tutorial
- EIGRP Tutorial
- OSPF Tutorial
- Practice CCNA GNS3 Labs
- CCNA Lab Challenges

CCNA 200-301

- CCNA FAQs & Tips
- Basic Questions
- Topology Architecture Questions
- Cloud & Virtualization Questions
- CDP & LLDP Questions
- Switch Questions
- VLAN & Trunking Questions
- VLAN & Trunking Questions 2
- STP & VTP Questions
- EtherChannel Questions
- TCP & UDP Questions
- IP Address & Subnetting Questions
- IP Routing Questions
- IP Routing Questions 2
- OSPF Questions
- OSPF Questions 2
- EIGRP Questions
- NAT Questions
- NTP Questions
- Syslog Questions

- HSRP Questions
- Access-list Questions
- AAA Questions
- Security Questions
- Security Questions 2
- DAI Questions
- IPv6 Questions
- DNS Questions
- QoS Questions
- Port Security Questions
- Wireless Questions
- Wireless Questions 2
- SDN Questions
- DNA Center Questions
- Drag Drop Questions
- Drag Drop Questions 2
- Drag Drop Questions 3
- VPN Questions
- DHCP Questions
- Automation Questions
- Miscellaneous Questions
- Share your CCNA Experience

Network Resources

- Free Router Simulators
- CCNA Website
- ENCOR Website
- ENSDWI Website
- ENARSI Website
- DevNet Website
- CCIE R&S Website
- Security Website
- Wireless Website
- Design Website
- Data Center Website
- Service Provider Website
- Collaboration Website

Top



Copyright © 2014 - 2026 CCNA Training
Privacy Policy | Disclaimer | Contact Us